

PHASE 3: PROJECT DESIGN

1. System Architecture

LegalEase follows a simple architecture consisting of three major components:

1. **Frontend – Streamlit**
2. **Backend – FastAPI**
3. **AI Core – Google Gemini**

The frontend collects the required information from the user. The FastAPI backend receives the request and communicates with the AI component. Gemini generates the legal document content, which is then returned to the frontend for preview, editing, and downloading.

2. Frontend Design

The LegalEase frontend is designed using Streamlit.

The interface provides input fields for:

- Document Type
- Parties Involved
- Terms and Conditions
- Effective Date

It also provides options to:

- Generate Document
- Edit Document
- Preview Generated Content
- Download the Document

3. Backend Design

The backend is developed using FastAPI.

The backend provides a /generate endpoint that accepts the user's document requirements and processes the request through the Gemini document generator.

4. AI Integration Design

LegalEase integrates Google Gemini for AI-powered legal document generation.

The AI component receives structured information from the backend, including:

- Document type
- Parties involved
- Terms and conditions
- Effective date

The Gemini document generator processes these inputs and returns the generated legal content to the FastAPI backend.

5. Document Generation Design

The generated content is processed into different output formats.

TXT

Provides the generated document as plain text.

DOCX

Uses the python-docx library to create a formatted Word document. It can include the logo, headings, terms table, and footer.

PDF

Uses FPDF to create a formatted PDF with a logo, headings, bullet-style terms, header, and footer.

HTML Preview

The generated content can also be displayed as a styled HTML preview within the Streamlit application.

6. User Flow

The basic user flow is:

Enter Details → Generate Document → AI Processing → Preview → Edit → Download

This design provides a simple workflow from entering the required information to obtaining the final legal document.

7. Output and Formatting Design

LegalEase supports multiple document output formats based on the generated legal content.

- **TXT:** Plain text version of the generated document.
- **DOCX:** Editable Word document with formatting, logo, headings, terms table, and footer.
- **PDF:** Formatted document with logo, headings, bullet-style terms, header, and footer.
- **HTML Preview:** Displays the generated document in a styled preview within the application.

8. Editing Design

After generation, the document content can be displayed in an editable area. Users can modify the generated text before downloading the final document.

This allows users to review and customize the generated legal content.

9. Overall Design Flow

The overall system design follows this sequence:

User Input → Streamlit Frontend → FastAPI Backend → Gemini AI → Generated Document → Preview/Edit → Download

The design connects the user interface, backend processing, AI generation, and document formatting components into one workflow.

10. Conclusion

The Project Design provides a structured architecture for LegalEase by combining Streamlit, FastAPI, Gemini AI, and document formatting modules. The design supports generation, preview, editing, and downloading of legal documents in different formats.